

Relational Database Schema & Complete Data Dictionary

Comprehensive technical schema documentation covering all 17 database tables, primary/foreign key relationships, indexing strategies, field specifications, and sample records.

System Name	Healthy Bite – Digital Restaurant & Food Ordering System
Database Name	healthy_bite
Database Engine	MySQL / MariaDB (InnoDB)
Default Collation	utf8mb4_unicode_ci
Total Tables	17 Base Tables
Total Seeded Records	346 Records (Base Seeder) / 508 Records (Live Database)
Generated At	2026-10-02 08:37:06 IST (Schema Verified)
Architecture	Custom MVC (PDO, Strict Typing, RESTful API)

About this Specification:
The Healthy Bite relational database is structured to support a high-performance digital restaurant ecosystem featuring multi-tenant restaurant management, dynamic dietary menu items with real-time macronutrient calculation (calories in kcal, protein, carbs, fat, fiber, sugar in grams, and sodium in milligrams), granular food variant portions, multi-group food customization, dine-in QR table ordering sessions, immutable snapshot audit trails for placed orders, and end-to-end cart order processing.

1. Database Tables Directory

Overview of all 17 relational tables organized alphabetically with domain module, row count, and primary purpose. Total baseline seed count: **346 records** (508 records in active operational instance).

#	Table Name	Domain Module	Rows	Purpose & Description
1	admin	Authentication & Access Control	4	Administrator portal accounts for platform management and global configuration.
2	branches	Restaurant & Infrastructure	1	Restaurant physical branch locations with contact information and operation status.
3	categories	Food Menu & Catalog	8	Menu categories (Bowls, Wraps, Beverages, Desserts, etc.) with display sort ordering.
4	customers	Orders & Commerce	12	Registered and guest customer profiles with mobile numbers, names, and contact details.
5	food_customizations	Food Menu & Catalog	129	Meal customization options (bases, proteins, sauces, toppings) with macro and price adjustments.
6	food_items	Food Menu & Catalog	45	Menu items/meals with full nutritional breakdown (calories, protein, carbs, fat), ingredients, and photos.
7	food_variants	Food Menu & Catalog	76	Meal portion and size variants (e.g. Standard vs Large, 8-inch vs 11-inch) with macro and price adjustments.
8	order_item_customizations	Orders & Commerce	19	Customization choices attached to specific line items in customer orders.
9	order_items	Orders & Commerce	13	Individual ordered food items with line total, quantity, and snapshot nutrition metrics.
10	orders	Orders & Commerce	10	Customer orders with subtotal, 5% GST, optional service charges, total, lifecycle status, and table reference.
11	payments	Orders & Commerce	6	Payment transaction records including payment methods, status, transaction IDs, and timestamps.
12	qr_tokens	Restaurant & Infrastructure	1	QR table ordering session tokens with expiration and active status validation.
13	restaurant_tables	Restaurant & Infrastructure	4	Dine-in restaurant tables associated with branches and active occupancy status.
14	restaurants	Restaurant & Infrastructure	1	Multi-tenant restaurant entity details, branding, logos, cover photos, and owner assignments.
15	reviews	Feedback & Quality	0	Customer feedback ratings, overall dining reviews, and restaurant owner reply tracking.
16	roles	Authentication & Access Control	4	User authorization roles (Super Admin, Restaurant Owner, Manager, Staff).
17	users	Authentication & Access Control	3	Staff and owner portal user accounts with encrypted passwords and role assignments.

2. Detailed Table Specifications

Complete data dictionary for each table containing column specifications, data types, nullability, keys, default values, foreign key constraints, composite unique indexes, architecture advisories, and representative sample data.

1. admin [Authentication & Access Control] (4 records · InnoDB)

Purpose: Administrator portal accounts for platform management and global configuration.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
name	varchar(120)	NO	—	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Architecture Advisory: The admin table is preserved for platform auditing records. Active platform administrator authentication, session handling, and access control are dynamically handled via the users table with role_id = 1 (Super Admin). Refer to users and roles specifications.

Sample Records Preview:

id	name	created_at	updated_at
1	Platform Super Admin	2026-08-30 15:50:38	2026-08-30 15:50:38
2	Tenant Administrator	2026-08-30 15:50:38	2026-08-30 15:50:38

2. branches [Restaurant & Infrastructure] (1 records · InnoDB)

Purpose: Restaurant physical branch locations with contact information and operation status.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
name	varchar(150)	NO	—	NULL	—
address	text	YES	—	NULL	—
city	varchar(100)	YES	—	NULL	—
state	varchar(100)	YES	—	NULL	—
phone	varchar(30)	YES	—	NULL	—
status	enum('active','inactive')	NO	IDX	'active'	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_branches_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE

Sample Records Preview:

id	restaurant_id	name	address	city	state
1	1	Greenhouse Kitchen – Indiranagar	100 Feet Road, HAL 2nd Stage, Indiranagar	Bengaluru	Karnataka

3. categories [Food Menu & Catalog] (8 records · InnoDB)

Purpose: Menu categories (Bowls, Wraps, Beverages, Desserts, Salads, Soups, etc.) with display sort ordering.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
name	varchar(100)	NO	—	NULL	—
slug	varchar(100)	NO	IDX	NULL	—
description	text	YES	—	NULL	—
image	varchar(255)	YES	—	NULL	—
sort_order	int(11)	NO	—	0	—
status	enum('active','inactive')	NO	—	'active'	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_categories_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE

Sample Records Preview:

id	restaurant_id	name	slug	description	image
1	1	Bowls	bowls	Nutrient-packed warm and cold signature bowls	cat_bowls.jpg
2	1	Wraps	wraps	Whole-wheat high-protein roll-ups	cat_wraps.jpg

4. customers [Orders & Commerce] (12 records · InnoDB)

Purpose: Registered and guest customer profiles with mobile numbers, names, and contact details.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
name	varchar(120)	NO	—	NULL	—
mobile	varchar(30)	YES	IDX	NULL	—
email	varchar(191)	YES	IDX	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Architecture Advisory: Design Note: Customers are created during checkout or QR ordering. Mobile and email fields are nullable to accommodate fast guest dining checkouts while maintaining indexed lookups for repeat customer profiles.

Sample Records Preview:

id	name	mobile	email	created_at	updated_at
1	Rahul Sharma	+919876543210	rahul.sharma@example.com	2026-09-14 14:17:24	2026-09-14 14:17:24
2	Rohan Varma	+919988776655	—	2026-09-14 14:27:57	2026-09-14 14:27:57

5. food_customizations [Food Menu & Catalog] (129 records · InnoDB)

Purpose: Meal customization options (bases, proteins, sauces, toppings) with macro nutrition and price adjustments.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
food_item_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
name	varchar(100)	NO	—	NULL	—
description	varchar(255)	YES	—	NULL	—
group_name	varchar(100)	NO	—	NULL	—
price_adjustment	decimal(10,2)	NO	—	0.00	—
calories_adjustment	int(11)	YES	—	NULL	—
protein_adjustment	decimal(6,2)	YES	—	NULL	—
carbs_adjustment	decimal(6,2)	YES	—	NULL	—
fat_adjustment	decimal(6,2)	YES	—	NULL	—
fiber_adjustment	decimal(6,2)	YES	—	NULL	—
sugar_adjustment	decimal(6,2)	YES	—	NULL	—
sodium_adjustment	decimal(7,2)	YES	—	NULL	—
caffeine_adjustment	decimal(6,2)	YES	—	NULL	—
is_required	tinyint(1)	NO	—	0	—
min_quantity	int(10) unsigned	NO	—	0	—
max_quantity	int(10) unsigned	NO	—	1	—
is_available	tinyint(1)	NO	IDX	1	—
sort_order	int(11)	NO	—	0	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_food_customizations_food_item	food_item_id	food_items.id	CASCADE	CASCADE

Architecture Advisory: Nutritional Units: Macronutrients (protein, carbs, fat, fiber, sugar, caffeine) are in grams (g), calories in kcal (signed int allows negative swaps), and sodium in milligrams (mg) using decimal(7,2).

Sample Records Preview:

id	food_item_id	name	description	group_name	price_adjustment
11	13	Extra Protein Scoop / Serving	—	Portion Add-on	40.00
12	13	Avocado Slices	—	Portion Add-on	50.00

6. food_items [Food Menu & Catalog] (45 records · InnoDB)

Purpose: Menu items / meals with full nutritional breakdown (calories, protein, carbs, fat, sugar), ingredients, allergens, and photos.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
category_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
name	varchar(150)	NO	—	NULL	—
slug	varchar(150)	NO	IDX	NULL	—
description	text	YES	—	NULL	—
image	varchar(255)	YES	—	NULL	—
ingredients	text	YES	—	NULL	—
allergens	varchar(255)	YES	—	NULL	—
food_type	enum('vegetarian','non_vegetarian','vegan','jain','other')	NO	—	'vegetarian'	—
base_price	decimal(10,2)	NO	—	NULL	—
calories	int(10) unsigned	YES	—	NULL	—
protein	decimal(6,2)	YES	—	NULL	—
carbs	decimal(6,2)	YES	—	NULL	—
fat	decimal(6,2)	YES	—	NULL	—
fiber	decimal(6,2)	YES	—	NULL	—
sugar	decimal(6,2)	YES	—	NULL	—
sodium	decimal(7,2)	YES	—	NULL	—
caffeine	decimal(6,2)	YES	—	NULL	—
serving_size	varchar(100)	YES	—	NULL	—
is_featured	tinyint(1)	NO	IDX	0	—
is_popular	tinyint(1)	NO	IDX	0	—
is_available	tinyint(1)	NO	IDX	1	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_food_items_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE
fk_food_items_category	category_id	categories.id	RESTRICT	CASCADE

Sample Records Preview:

id	restaurant_id	category_id	name	slug	description
1	1	1	Paneer Protein Bowl	paneer-protein-bowl	Tandoori paneer, brown rice, fresh veggies
2	1	1	Chicken Rice Bowl	chicken-rice-bowl	Grilled sous-vide chicken breast with greens

7. food_variants [Food Menu & Catalog] (76 records · InnoDB)

Purpose: Meal portion and size variants (e.g. Standard vs Large, 8-inch vs 11-inch) with macro and price adjustments.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
food_item_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
name	varchar(100)	NO	—	NULL	—
description	varchar(255)	YES	—	NULL	—
price_adjustment	decimal(10,2)	NO	—	0.00	—
calories_adjustment	int(11)	YES	—	NULL	—
protein_adjustment	decimal(6,2)	YES	—	NULL	—
carbs_adjustment	decimal(6,2)	YES	—	NULL	—
fat_adjustment	decimal(6,2)	YES	—	NULL	—
fiber_adjustment	decimal(6,2)	YES	—	NULL	—
sugar_adjustment	decimal(6,2)	YES	—	NULL	—
sodium_adjustment	decimal(7,2)	YES	—	NULL	—
caffeine_adjustment	decimal(6,2)	YES	—	NULL	—
is_required	tinyint(1)	NO	—	0	—
is_available	tinyint(1)	NO	IDX	1	—
sort_order	int(11)	NO	—	0	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints & Compound Indexes:

Constraint Name	Column	References	On Delete	On Update
fk_food_variants_food_item	food_item_id	food_items.id	CASCADE	CASCADE

Architecture Advisory: Indexing Strategy: Compound index idx_food_variants_available (food_item_id, is_available) enables instant scoped querying during menu render.

Sample Records Preview:

id	food_item_id	name	description	price_adj	calories_adj
5	13	Standard Portion	—	0.00	0
6	13	Large / Power Portion	—	50.00	110

8. order_item_customizations [Orders & Commerce] (19 records · InnoDB)

Purpose: Customization choices attached to specific line items in customer orders.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
order_item_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
customization_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
customization_name_snapsho t	varchar(100)	NO	—	NULL	—
quantity	int(10) unsigned	NO	—	1	—
price_adjustment	decimal(10,2)	NO	—	0.00	—
calories_adjustment	int(11)	YES	—	NULL	—
protein_adjustment	decimal(6,2)	YES	—	NULL	—
carbs_adjustment	decimal(6,2)	YES	—	NULL	—
fat_adjustment	decimal(6,2)	YES	—	NULL	—
fiber_adjustment	decimal(6,2)	YES	—	NULL	—
sugar_adjustment	decimal(6,2)	YES	—	NULL	—
sodium_adjustment	decimal(7,2)	YES	—	NULL	—
caffeine_adjustment	decimal(6,2)	YES	—	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_order_custom_item	order_item_id	order_items.id	CASCADE	CASCADE
fk_order_custom_def	customization_id	food_customizations.id	RESTRICT	CASCADE

Architecture Advisory: Historical Immutability: Customization names, prices, and nutritional impacts are stored as snapshots to guarantee historical order reporting integrity.

Sample Records Preview:

id	order_item_id	customization_id	name_snapshot	qty	price_adj
1	1	1	Extra Paneer	1	40.00
2	1	2	Avocado	1	60.00

9. order_items [Orders & Commerce] (13 records · InnoDB)

Purpose: Individual ordered food items with line total, quantity, and snapshot nutrition metrics.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
order_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
food_item_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
food_name_snapshot	varchar(150)	NO	—	NULL	—
base_price_snapshot	decimal(10,2)	NO	—	NULL	—
variant_name_snapshot	varchar(100)	YES	—	NULL	—
variant_price_snapshot	decimal(10,2)	NO	—	0.00	—
quantity	int(10) unsigned	NO	—	1	—
unit_price	decimal(10,2)	NO	—	NULL	—
total_price	decimal(10,2)	NO	—	NULL	—
calories	int(11)	YES	—	NULL	—
protein	decimal(6,2)	YES	—	NULL	—
carbs	decimal(6,2)	YES	—	NULL	—
fat	decimal(6,2)	YES	—	NULL	—
fiber	decimal(6,2)	YES	—	NULL	—
sugar	decimal(6,2)	YES	—	NULL	—
sodium	decimal(7,2)	YES	—	NULL	—
caffeine	decimal(6,2)	YES	—	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_order_items_order	order_id	orders.id	RESTRICT	CASCADE
fk_order_items_food	food_item_id	food_items.id	RESTRICT	CASCADE

Sample Records Preview:

id	order_id	food_item_id	food_name_snapshot	base_price_snapshot	variant_name_snapshot
1	1	1	Paneer Protein Bowl	249.00	Large
2	2	1	Paneer Protein Bowl	249.00	Large

10. orders [Orders & Commerce] (10 records · InnoDB)

Purpose: Customer orders with subtotal, tax (5% GST), optional service charge, total pricing, status lifecycle, and table reference.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
order_number	varchar(64)	NO	UNI	NULL	—
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
branch_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
table_id	bigint(20) unsigned	YES	FK/IDX	NULL	—
customer_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
order_type	enum('dine_in','takeaway')	NO	—	'dine_in'	—
subtotal	decimal(10,2)	NO	—	0.00	—
tax	decimal(10,2)	NO	—	0.00	—
service_charge	decimal(10,2)	NO	—	0.00	—
total_amount	decimal(10,2)	NO	—	0.00	—
payment_status	enum('pending','completed','failed')	NO	IDX	'pending'	—
order_status	enum('placed','accepted','preparing','ready','completed','cancelled')	NO	IDX	'placed'	—
notes	text	YES	—	NULL	—
created_at	timestamp	NO	IDX	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_orders_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE
fk_orders_branch	branch_id	branches.id	RESTRICT	CASCADE
fk_orders_table	table_id	restaurant_tables.id	SET NULL	CASCADE
fk_orders_customer	customer_id	customers.id	RESTRICT	CASCADE

Sample Records Preview:

id	order_number	restaurant_id	branch_id	table_id	customer_id
1	HB-1A09F232DD9-412	1	1	4	2
2	HB-1A09F2BF467-645	1	1	4	3

11. payments [Orders & Commerce] (6 records · InnoDB)

Purpose: Payment transaction settlement records including payment methods, status, transaction IDs, and timestamps.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
order_id	bigint(20) unsigned	NO	UNI	NULL	—
payment_method	enum('cash','upi','card')	NO	—	NULL	—
transaction_reference	varchar(128)	YES	—	NULL	—
amount	decimal(10,2)	NO	—	NULL	—
status	enum('pending','completed','failed')	NO	IDX	'pending'	—
paid_at	timestamp	YES	—	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_payments_order	order_id	orders.id	RESTRICT	CASCADE

Architecture Advisory: Settlement Constraint Note: order_id is marked UNIQUE to enforce 1-to-1 order settlement. For split payments or multi-attempt retry logs, relaxing UNIQUE to an INDEX is recommended.

Sample Records Preview:

id	order_id	method	transaction_ref	amount	status
1	1	upi	TXN-UPI-8CBFCF82D856	837.90	completed
2	2	upi	TXN-UPI-14FDE3566DEC	837.90	completed

12. qr_tokens [Restaurant & Infrastructure] (1 records · InnoDB)

Purpose: QR table ordering session tokens with expiration and active status validation.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
branch_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
table_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
token	varchar(128)	NO	UNI	NULL	—
status	enum('active','inactive','expired')	NO	IDX	'active'	—
expires_at	timestamp	YES	—	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_qr_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE
fk_qr_branch	branch_id	branches.id	RESTRICT	CASCADE
fk_qr_table	table_id	restaurant_tables.id	RESTRICT	CASCADE

Sample Records Preview:

id	restaurant_id	branch_id	table_id	token	status
1	1	1	4	hb_qr_token_indiranagar_tbl4_7f8a9b	active

13. restaurant_tables [Restaurant & Infrastructure] (4 records · InnoDB)

Purpose: Dine-in restaurant tables associated with branches and active occupancy status.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
branch_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
table_number	varchar(50)	NO	UNI*	NULL	—
status	enum('available','occupied','cleaning','out_of_service')	NO	IDX	'available'	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints & Unique Index:

Constraint Name	Column	References	On Delete	On Update
fk_tables_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE
fk_tables_branch	branch_id	branches.id	RESTRICT	CASCADE

Architecture Advisory: Composite Unique Key: uq_branch_table (branch_id, table_number) prevents duplicate table numbering within the same physical branch.

Sample Records Preview:

id	restaurant_id	branch_id	table_number	status	created_at
1	1	1	Table 1	available	2026-09-14 14:17:24
2	1	1	Table 2	available	2026-09-14 14:17:24

14. restaurants [Restaurant & Infrastructure] (1 records · InnoDB)

Purpose: Multi-tenant restaurant entity details, branding, logos, cover photos, and owner assignments.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
owner_user_id	bigint(20) unsigned	YES	FK/IDX	NULL	—
name	varchar(150)	NO	—	NULL	—
slug	varchar(150)	NO	UNI	NULL	—
logo	varchar(255)	YES	—	NULL	—
cover_image	varchar(255)	YES	—	NULL	—
description	text	YES	—	NULL	—
phone	varchar(30)	YES	—	NULL	—
email	varchar(191)	YES	—	NULL	—
address	text	YES	—	NULL	—
city	varchar(100)	YES	—	NULL	—
state	varchar(100)	YES	—	NULL	—
status	enum('pending','approved','suspended')	NO	IDX	'approved'	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_restaurants_owner_user	owner_user_id	users.id	SET NULL	CASCADE

Architecture Advisory: Circular Relationship Resolution: restaurants.owner_user_id links to users.id, while users.restaurant_id links to restaurants.id. The restaurant is seeded with owner_user_id = NULL, the user created, and then linked via UPDATE.

Sample Records Preview:

id	owner_user_id	name	slug	logo	cover_image
1	2	Greenhouse Kitchen	greenhouse-kitchen	greenhouse_logo.png	greenhouse_cover.jpg

15. reviews [Feedback & Quality] (0 records · InnoDB)

Purpose: Customer feedback ratings, reviews on dining experience, and restaurant management reply tracking.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
restaurant_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
customer_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
order_id	bigint(20) unsigned	YES	FK/IDX	NULL	—
rating	tinyint(3) unsigned	NO	—	NULL	—
comment	text	YES	—	NULL	—
restaurant_reply	text	YES	—	NULL	—
replied_at	timestamp	YES	—	NULL	—
status	enum('pending','approved','hidden')	NO	IDX	'pending'	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_reviews_order	order_id	orders.id	SET NULL	CASCADE
fk_reviews_customer	customer_id	customers.id	RESTRICT	CASCADE
fk_reviews_restaurant	restaurant_id	restaurants.id	RESTRICT	CASCADE

Architecture Advisory: Review Replies Support: Synchronized with migration add_review_replies.php, adding restaurant_reply and replied_at to allow verified restaurant owners to respond to feedback.

Sample Records Preview:

id	restaurant_id	customer_id	order_id	rating	comment	restaurant_reply
1	1	1	1	5	Fresh food, fantastic taste!	Thank you!

16. roles [Authentication & Access Control] (4 records · InnoDB)

Purpose: User authorization roles (Super Admin, Restaurant Owner, Manager, Staff).

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
name	varchar(50)	NO	—	NULL	—
slug	varchar(50)	NO	UNI	NULL	—
description	varchar(255)	YES	—	NULL	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Sample Records Preview:

id	name	slug	description	created_at
1	Super Admin	super_admin	Platform administrator with full privileges	2026-09-14 14:17:24
2	Restaurant Owner	restaurant_owner	Owner of a restaurant entity	2026-09-14 14:17:24

17. users [Authentication & Access Control] (3 records · InnoDB)

Purpose: Staff and owner portal user accounts with encrypted passwords and role assignments.

Column Name	Type	Null	Key	Default	Extra
id	bigint(20) unsigned	NO	PK	NULL	auto_increment
role_id	bigint(20) unsigned	NO	FK/IDX	NULL	—
restaurant_id	bigint(20) unsigned	YES	FK/IDX	NULL	—
name	varchar(120)	NO	—	NULL	—
email	varchar(191)	NO	UNI	NULL	—
password	varchar(255)	NO	—	NULL	—
status	enum('active','inactive','suspended')	NO	IDX	'active'	—
created_at	timestamp	NO	—	current_timestamp()	—
updated_at	timestamp	NO	—	current_timestamp()	on update current_timestamp()

Foreign Key Constraints:

Constraint Name	Column	References	On Delete	On Update
fk_users_role	role_id	roles.id	RESTRICT	CASCADE
fk_users_restaurant	restaurant_id	restaurants.id	SET NULL	CASCADE

Architecture Advisory: Super Admin Scoping: Users with role_id = 1 (Super Admin) have restaurant_id = NULL, giving them unconstrained platform-wide access across all tenant restaurants.

Sample Records Preview:

id	role_id	restaurant_id	name	email	password
1	1	—	System Admin	admin@healthybite.com	\$2y\$12\$***** (bcrypt)
2	2	1	Aarav Patel	owner@greenhousekitchen.com	\$2y\$12\$***** (bcrypt)